PHYS4055

Atomic physics and quantum optics

Week 8

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In these lectures, we will look into:

- Filling of electron shells
- Coupling of electrons, term nomenclature
- Wigner-Eckart theorem, 3-j and 6-j symbols, spherical tensor operators
- Hartree-Fock approximation
- Dirac equation

Spectra of multi-electron atoms

Consider central field V(r).

For H-like, $V(r) = -Z/r$.

⇒ many-electron system, electrons screen the Coulomb field of nucleus, $V(r) = -Z/r + V_{el}(r)$ (neglecting e-e interactions)

Ordering/filling of shells.

Fermions ⟹ Pauli exclusion principle.

How many e's in each shell?

Can have $2 \times (2\ell+1)$ in each shell

- 2: spin (up, down)
- $(2\ell+1)$: projections m_ℓ for each ℓ.

$\ell =$	0	1	2	3	...
$2(2\ell+1) =$	2	6	10	14	...

$1s^2 2s^2 2p^6 \ldots$

Spectra of multi-electron atoms

PERIODIC TABLE

Atomic Properties of the Elements

FREQUENTLY USED FUNDAMENTAL PHYSICAL CONSTANTS[§]

1 second = 9 192 631 770 periods of radiation corresponding to the transition between the two hyperfine levels of the ground state of ^{133}Cs

speed of light in vacuum	c	299 792 458 m s^{-1}	(exact)
Planck constant	h	6.626 070 15 x 10^{-34} J Hz^{-1}	(exact)
elementary charge	e	1.602 176 634 x 10^{-19} C	(exact)
Avogadro constant	N_A	6.022 140 76 x 10^{23} mol^{-1}	(exact)
Boltzmann constant	k	1.380 649 x 10^{-23} J K^{-1}	(exact)
electron volt	eV	1.602 176 634 x 10^{-19} J	(exact)
electron mass	m_e	9.109 383 70 x 10^{-31} kg	
energy equivalent	$m_e c^2$	0.510 998 950 MeV	
proton mass	m_p	1.672 621 924 x 10^{-27} kg	
energy equivalent	$m_p c^2$	938.272 088 MeV	
fine-structure constant	α	1/137.035 999	
Rydberg energy	$R_\infty hc$	13.605 693 1230 eV	
Newtonian constant of gravitation	G	6.674 x 10^{-11} m^3 kg^{-1} s^{-2}	

[§] For the most accurate values of these and other constants, visit pml.nist.gov/constants.

Solids
Liquids
Gases
Artificially Prepared

National Institute of Standards and Technology
U.S. Department of Commerce

Physical Measurement Laboratory www.nist.gov/pml
Standard Reference Data www.nist.gov/srd

Period	Group 1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VIII	10	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 $^2S_{1/2}$ **H** Hydrogen 1.008 1s 13.5984																	2 $^1S_0$ **He** Helium 4.0026 1s^2 24.5874
2	3 $^2S_{1/2}$ **Li** Lithium 6.94 1s^{2}2s 5.3917	4 $^1S_0$ **Be** Beryllium 9.0122 1s^{2}2s^2 9.3227											5 $^2P^o_{1/2}$ **B** Boron 10.81 1s^{2}2s^{2}2p 8.2980	6 $^3P_0$ **C** Carbon 12.011 1s^{2}2s^{2}2p^2 11.2603	7 $^4S^o_{3/2}$ **N** Nitrogen 14.007 1s^{2}2s^{2}2p^3 14.5341	8 $^3P_2$ **O** Oxygen 15.999 1s^{2}2s^{2}2p^4 13.6181	9 $^2P^o_{3/2}$ **F** Fluorine 18.998 1s^{2}2s^{2}2p^5 17.4228	10 $^1S_0$ **Ne** Neon 20.180 1s^{2}2s^{2}2p^6 21.5645
3	11 $^2S_{1/2}$ **Na** Sodium 22.990 [Ne]3s 5.1391	12 $^1S_0$ **Mg** Magnesium 24.305 [Ne]3s^2 7.6462											13 $^2P^o_{1/2}$ **Al** Aluminum 26.982 [Ne]3s^{2}3p 5.9858	14 $^3P_0$ **Si** Silicon 28.085 [Ne]3s^{2}3p^2 8.1517	15 $^4S^o_{3/2}$ **P** Phosphorus 30.974 [Ne]3s^{2}3p^3 10.4867	16 $^3P_2$ **S** Sulfur 32.06 [Ne]3s^{2}3p^4 10.3600	17 $^2P^o_{3/2}$ **Cl** Chlorine 35.45 [Ne]3s^{2}3p^5 12.9676	18 $^1S_0$ **Ar** Argon 39.948 [Ne]3s^{2}3p^6 15.7596
4	19 $^2S_{1/2}$ **K** Potassium 39.098 [Ar]4s 4.3407	20 $^1S_0$ **Ca** Calcium 40.078 [Ar]4s^2 6.1132	21 $^2D_{3/2}$ **Sc** Scandium 44.956 [Ar]3d4s^2 6.5615	22 $^3F_2$ **Ti** Titanium 47.867 [Ar]3d^{2}4s^2 6.8281	23 $^4F_{3/2}$ **V** Vanadium 50.942 [Ar]3d^{3}4s^2 6.7462	24 $^7S_3$ **Cr** Chromium 51.996 [Ar]3d^{5}4s 6.7665	25 $^6S_{5/2}$ **Mn** Manganese 54.938 [Ar]3d^{5}4s^2 7.4340	26 $^5D_4$ **Fe** Iron 55.845 [Ar]3d^{6}4s^2 7.9025	27 $^4F_{9/2}$ **Co** Cobalt 58.933 [Ar]3d^{7}4s^2 7.8810	28 $^3F_4$ **Ni** Nickel 58.693 [Ar]3d^{8}4s^2 7.6399	29 $^2S_{1/2}$ **Cu** Copper 63.546 [Ar]3d^{10}4s 7.7264	30 $^1S_0$ **Zn** Zinc 65.38 [Ar]3d^{10}4s^2 9.3942	31 $^2P^o_{1/2}$ **Ga** Gallium 69.723 [Ar]3d^{10}4s^{2}4p 5.9993	32 $^3P_0$ **Ge** Germanium 72.630 [Ar]3d^{10}4s^{2}4p^2 7.8994	33 $^4S^o_{3/2}$ **As** Arsenic 74.922 [Ar]3d^{10}4s^{2}4p^3 9.7886	34 $^3P_2$ **Se** Selenium 78.971 [Ar]3d^{10}4s^{2}4p^4 9.7524	35 $^2P^o_{3/2}$ **Br** Bromine 79.904 [Ar]3d^{10}4s^{2}4p^5 11.8138	36 $^1S_0$ **Kr** Krypton 83.798 [Ar]3d^{10}4s^{2}4p^6 13.9996
5	37 $^2S_{1/2}$ **Rb** Rubidium 85.468 [Kr]5s 4.1771	38 $^1S_0$ **Sr** Strontium 87.62 [Kr]5s^2 5.6949	39 $^2D_{3/2}$ **Y** Yttrium 88.906 [Kr]4d5s^2 6.2173	40 $^3F_2$ **Zr** Zirconium 91.224 [Kr]4d^{2}5s^2 6.6341	41 $^6D_{1/2}$ **Nb** Niobium 92.906 [Kr]4d^{4}5s 6.7589	42 $^7S_3$ **Mo** Molybdenum 95.95 [Kr]4d^{5}5s 7.0924	43 $^6S_{5/2}$ **Tc** Technetium (97) [Kr]4d^{5}5s^2 7.1194	44 $^5F_5$ **Ru** Ruthenium 101.07 [Kr]4d^{7}5s 7.3605	45 $^4F_{9/2}$ **Rh** Rhodium 102.91 [Kr]4d^{8}5s 7.4589	46 $^1S_0$ **Pd** Palladium 106.42 [Kr]4d^{10} 8.3369	47 $^2S_{1/2}$ **Ag** Silver 107.87 [Kr]4d^{10}5s 7.5762	48 $^1S_0$ **Cd** Cadmium 112.41 [Kr]4d^{10}5s^2 8.9938	49 $^2P^o_{1/2}$ **In** Indium 114.82 [Kr]4d^{10}5s^{2}5p 5.7864	50 $^3P_0$ **Sn** Tin 118.71 [Kr]4d^{10}5s^{2}5p^2 7.3439	51 $^4S^o_{3/2}$ **Sb** Antimony 121.76 [Kr]4d^{10}5s^{2}5p^3 8.6084	52 $^3P_2$ **Te** Tellurium 127.60 [Kr]4d^{10}5s^{2}5p^4 9.0097	53 $^2P^o_{3/2}$ **I** Iodine 126.90 [Kr]4d^{10}5s^{2}5p^5 10.4513	54 $^1S_0$ **Xe** Xenon 131.29 [Kr]4d^{10}5s^{2}5p^6 12.1298
6	55 $^2S_{1/2}$ **Cs** Cesium 132.91 [Xe]6s 3.8939	56 $^1S_0$ **Ba** Barium 137.33 [Xe]6s^2 5.2117		72 $^3F_2$ **Hf** Hafnium 178.49 [Xe]4f^{14}5d^{2}6s^2 6.8251	73 $^4F_{3/2}$ **Ta** Tantalum 180.95 [Xe]4f^{14}5d^{3}6s^2 7.5496	74 $^5D_0$ **W** Tungsten 183.84 [Xe]4f^{14}5d^{4}6s^2 7.8640	75 $^6S_{5/2}$ **Re** Rhenium 186.21 [Xe]4f^{14}5d^{5}6s^2 7.8335	76 $^5D_4$ **Os** Osmium 190.23 [Xe]4f^{14}5d^{6}6s^2 8.4382	77 $^4F_{9/2}$ **Ir** Iridium 192.22 [Xe]4f^{14}5d^{7}6s^2 8.9670	78 $^3D_3$ **Pt** Platinum 195.08 [Xe]4f^{14}5d^{9}6s 8.9588	79 $^2S_{1/2}$ **Au** Gold 196.97 [Xe]4f^{14}5d^{10}6s 9.2256	80 $^1S_0$ **Hg** Mercury 200.59 [Xe]4f^{14}5d^{10}6s^2 10.4375	81 $^2P^o_{1/2}$ **Tl** Thallium 204.38 [Hg]6p 6.1083	82 $^3P_0$ **Pb** Lead 207.2 [Hg]6p^2 7.4167	83 $^4S^o_{3/2}$ **Bi** Bismuth 208.98 [Hg]6p^3 7.2855	84 $^3P_2$ **Po** Polonium (209) [Hg]6p^4 8.414	85 $^2P^o_{3/2}$ **At** Astatine (210) [Hg]6p^5 9.3175	86 $^1S_0$ **Rn** Radon (222) [Hg]6p^6 10.7485
7	87 $^2S_{1/2}$ **Fr** Francium (223) [Rn]7s 4.0727	88 $^1S_0$ **Ra** Radium (226) [Rn]7s^2 5.2784		104 $^3F_2$ **Rf** Rutherfordium (267) [Rn]5f^{14}6d^{2}7s^2 6.02	105 $^4F_{3/2}$ **Db** Dubnium (268) [Rn]5f^{14}6d^{3}7s^2 6.8	106 ^{0} **Sg** Seaborgium (269) [Rn]5f^{14}6d^{4}7s^2 7.8	107 $^{5/2}$ **Bh** Bohrium (270) [Rn]5f^{14}6d^{5}7s^2 7.7	108 ^{4} **Hs** Hassium (269) [Rn]5f^{14}6d^{6}7s^2 7.6	109 **Mt** Meitnerium (278)	110 **Ds** Darmstadtium (281)	111 **Rg** Roentgenium (282)	112 **Cn** Copernicium (285)	113 **Nh** Nihonium (286)	114 **Fl** Flerovium (289)	115 **Mc** Moscovium (289)	116 **Lv** Livermorium (293)	117 **Ts** Tennessine (294)	118 **Og** Oganesson (294)

Lanthanides	57 $^2D_{3/2}$ **La** Lanthanum 138.91 [Xe]5d6s^2 5.5769	58 $^1G^o_4$ **Ce** Cerium 140.12 [Xe]4f5d6s^2 5.5386	59 $^4I^o_{9/2}$ **Pr** Praseodymium 140.91 [Xe]4f^{3}6s^2 5.4702	60 $^5I_4$ **Nd** Neodymium 144.24 [Xe]4f^{4}6s^2 5.5250	61 $^6H^o_{5/2}$ **Pm** Promethium (145) [Xe]4f^{5}6s^2 5.577	62 $^7F_0$ **Sm** Samarium 150.36 [Xe]4f^{6}6s^2 5.6437	63 $^8S^o_{7/2}$ **Eu** Europium 151.96 [Xe]4f^{7}6s^2 5.6704	64 $^9D^o_2$ **Gd** Gadolinium 157.25 [Xe]4f^{7}5d6s^2 6.1498	65 $^6H^o_{15/2}$ **Tb** Terbium 158.93 [Xe]4f^{9}6s^2 5.8638	66 $^5I_8$ **Dy** Dysprosium 162.50 [Xe]4f^{10}6s^2 5.9391	67 $^4I^o_{15/2}$ **Ho** Holmium 164.93 [Xe]4f^{11}6s^2 6.0215	68 $^3H_6$ **Er** Erbium 167.26 [Xe]4f^{12}6s^2 6.1077	69 $^2F^o_{7/2}$ **Tm** Thulium 168.93 [Xe]4f^{13}6s^2 6.1843	70 $^1S_0$ **Yb** Ytterbium 173.05 [Xe]4f^{14}6s^2 6.2542	71 $^2D_{3/2}$ **Lu** Lutetium 174.97 [Xe]4f^{14}5d6s^2 5.4259
Actinides	89 $^2D_{3/2}$ **Ac** Actinium (227) [Rn]6d7s^2 5.3802	90 $^3F_2$ **Th** Thorium 232.04 [Rn]6d^{2}7s^2 6.3067	91 $^4K_{11/2}$ **Pa** Protactinium 231.04 [Rn]5f^{2}6d7s^2 5.89	92 $^5L^o_6$ **U** Uranium 238.03 [Rn]5f^{3}6d7s^2 6.1941	93 $^6L_{11/2}$ **Np** Neptunium (237) [Rn]5f^{4}6d7s^2 6.2655	94 $^7F_0$ **Pu** Plutonium (244) [Rn]5f^{6}7s^2 6.0258	95 $^8S^o_{7/2}$ **Am** Americium (243) [Rn]5f^{7}7s^2 5.9738	96 $^9D^o_2$ **Cm** Curium (247) [Rn]5f^{7}6d7s^2 5.9914	97 $^6H^o_{15/2}$ **Bk** Berkelium (247) [Rn]5f^{9}7s^2 6.1978	98 $^5I_8$ **Cf** Californium (251) [Rn]5f^{10}7s^2 6.2817	99 $^4I^o_{15/2}$ **Es** Einsteinium (252) [Rn]5f^{11}7s^2 6.3676	100 $^3H_6$ **Fm** Fermium (257) [Rn]5f^{12}7s^2 6.50	101 $^2F^o_{7/2}$ **Md** Mendelevium (258) [Rn]5f^{13}7s^2 6.58	102 $^1S_0$ **No** Nobelium (259) [Rn]5f^{14}7s^2 6.66	103 $^2P^o_{1/2}$ **Lr** Lawrencium (266) [Rn]5f^{14}7s^{2}7p 4.96

Atomic Number — Ground State
Symbol
Name
Standard Atomic Weight[†](u)
Ground-state Configuration — Ionization Energy (eV)

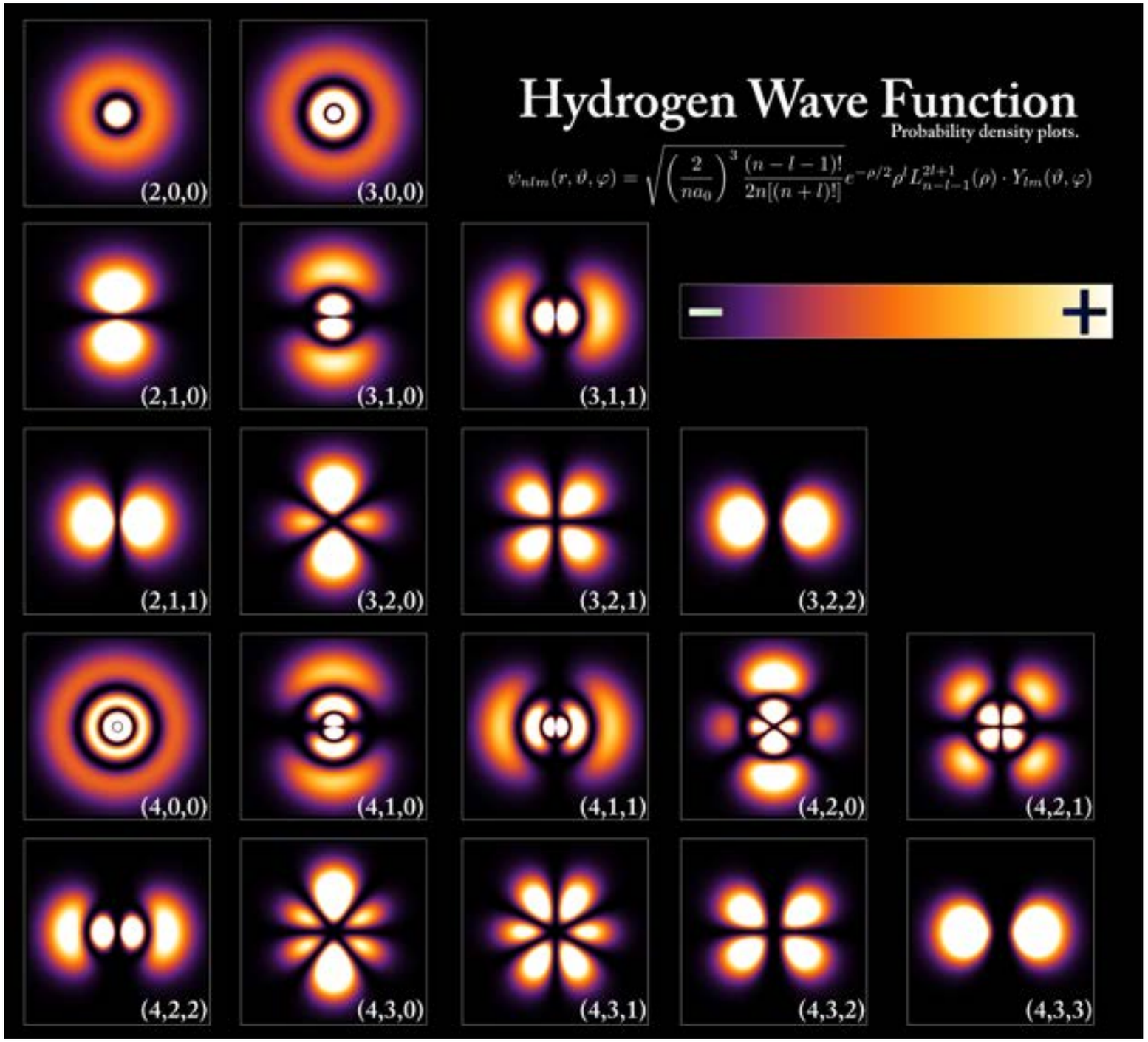
58 $^1G^o_4$
Ce
Cerium
140.12
[Xe]4f5d6s^2
5.5386

[†]Based upon ^{12}C. () indicates the mass number of the longest-lived isotope.

For the most precise values and uncertainties visit ciaaw.org and pml.nist.gov/data.

NIST SP 966 (July 2019)

Spectra of multi-electron atoms



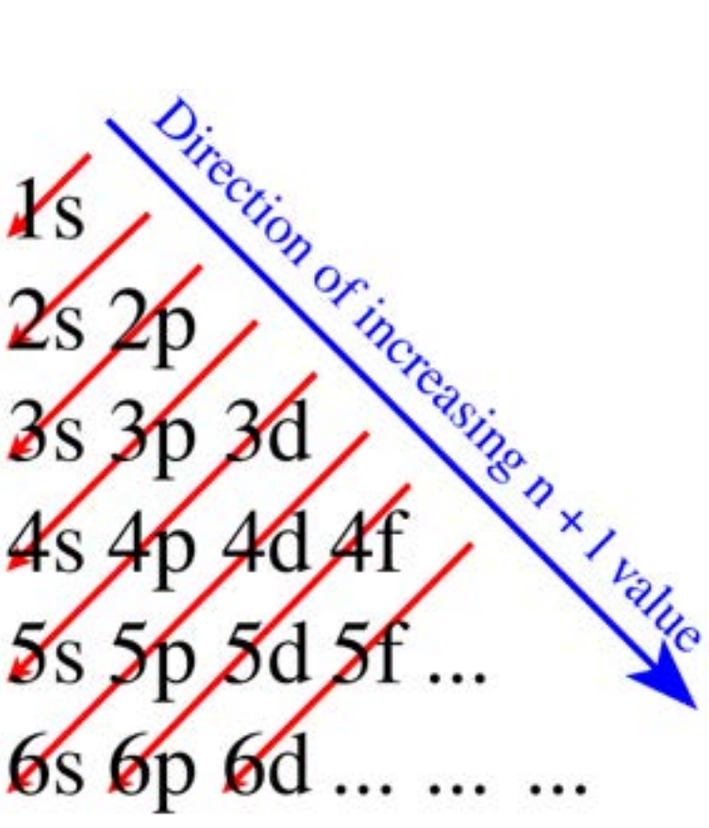
Wikipedia

Spectra of multi-electron atoms

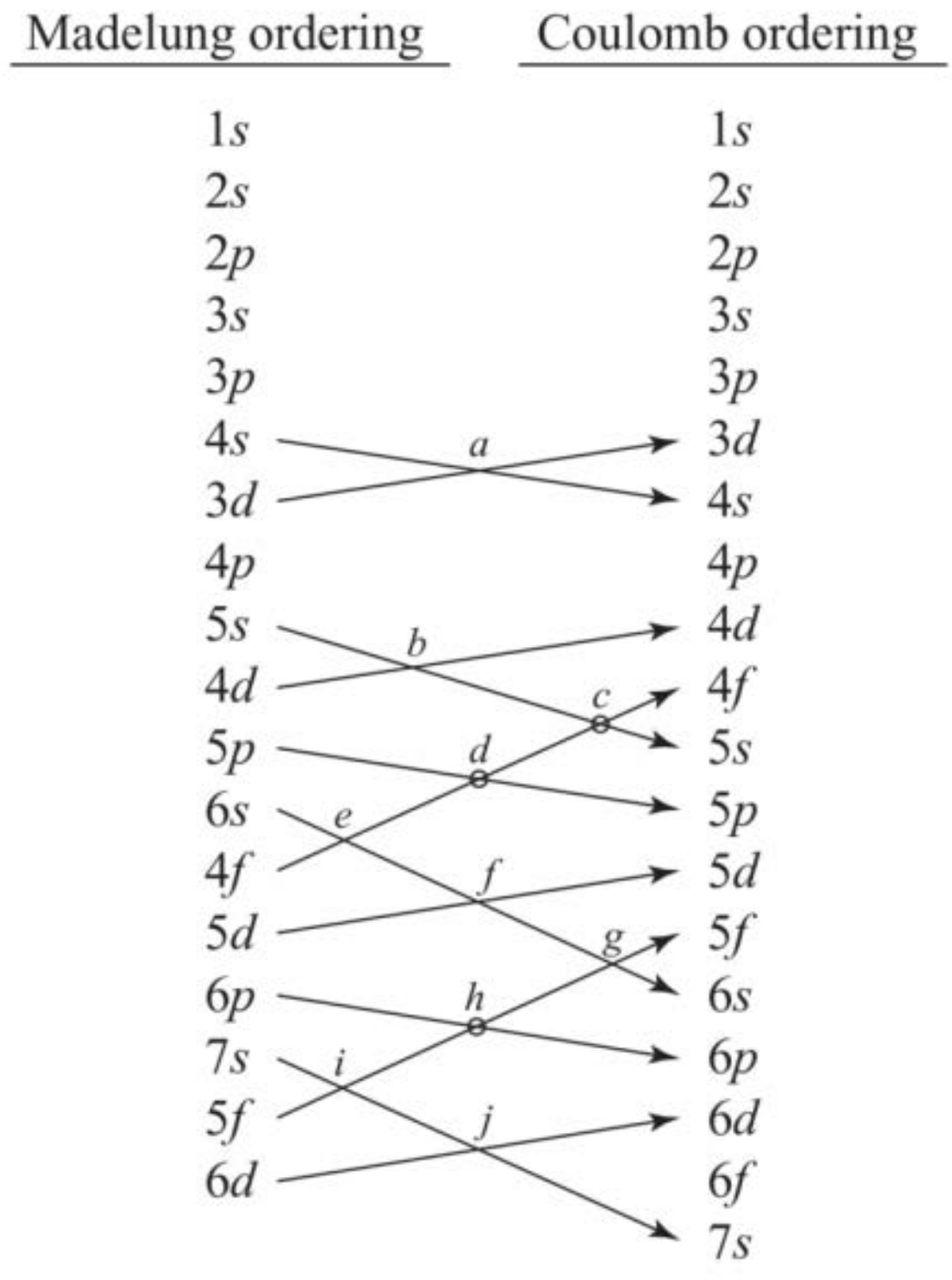
Ordering/filling of shells.

Atoms

States are filled in direction of red arrows. States crossed by same red line have the same value of *n+l.*



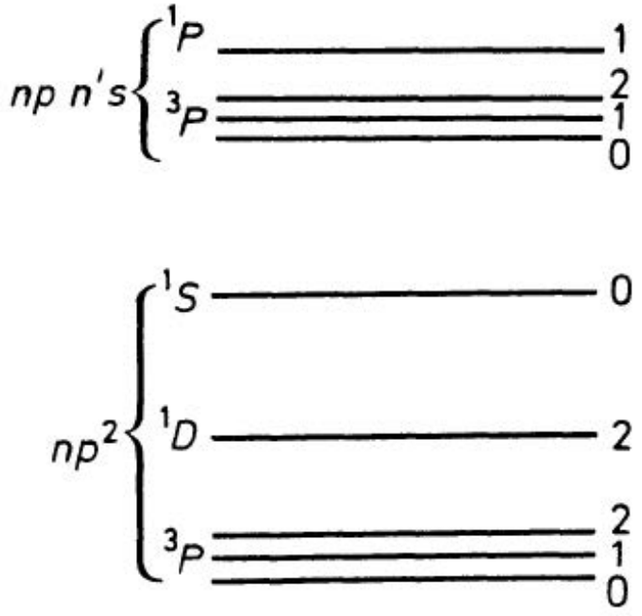
Wikipedia



Levels may cross when atoms are stripped of electrons, making them more Coulomb-like — this makes *optical highly-charged ion clocks* possible.

Spectra of multi-electron atoms

Coupling of angular momenta and nomenclature.



LS coupling of states

Exercise

(1) n p n's
↿⇂ ↿⇂
$l=1$ $l=0$
$s=\frac{1}{2}$ $s=\frac{1}{2}$

Spin: $\frac{1}{2}+\frac{1}{2}$, $\frac{1}{2}-\frac{1}{2}$

$|s_1 - s_2| \leq S \leq s_1 + s_2$

$\underset{\sim}{S} = \underset{\sim}{s}_1 + \underset{\sim}{s}_2$

$\Rightarrow \quad S = 1, 0.$

Orbital: $\underset{\sim}{L} = \underset{\sim}{l}_1 + \underset{\sim}{l}_2$

$\Rightarrow |l_1 - l_2| \leq L \leq l_1 + l_2$

$\Rightarrow \quad L = 1.$

n p n's terms:

$S=1, L=1 \Rightarrow J=2,1,0$

$S=0, L=1 \Rightarrow J=1$

$^3P_0, \ ^3P_1, \ ^3P_2$

$^1P_1$

Configuration: n p n's

Term: $^{(2S+1)}L_J$ ← total angular momentum, $\underset{\sim}{J} = \underset{\sim}{L} + \underset{\sim}{S}$

(2S+1) ← multiplicity

L ← orbital angular momentum.

$L = 0, 1, 2, 3, \ldots$
S P D F ...

$|L-S| \leq J \leq |L+S|$.

Note on multiplicity:
If $L \geq S$, splits into $2S+1$ states.
If $L < S$, splits into $2L+1$.

Notes: Hund's rule: State with max spin S has lowest energy.
Then, for set S, state with max L has lowest energy.

Here we used LS or Russell-Saunders coupling.

Here, the spins are first coupled together to obtain a total spin $\underset{\sim}{S}$,

$$\underset{\sim}{S}_1 + \underset{\sim}{S}_2 + \underset{\sim}{S}_3 + \cdots \Rightarrow \underset{\sim}{S}.$$

And separately, the orbital angular momenta are added,

$$\underset{\sim}{L}_1 + \underset{\sim}{L}_2 + \underset{\sim}{L}_3 + \cdots \Rightarrow \underset{\sim}{L}.$$

This is valid when the residual Coulomb interaction between electrons exceeds the spin-orbit interaction, most appropriate for light atoms.

$\Rightarrow$ $\underset{\sim}{L}$ & $\underset{\sim}{S}$ are then coupled together to obtain $\underset{\sim}{J}$

In jj coupling, the $\underset{\sim}{l}$ & $\underset{\sim}{s}$ of each electron are first combined to a $\underset{\sim}{j}$, & then the $\underset{\sim}{j}$'s of each electron are combined to a total $\underset{\sim}{J}$, $\underset{\sim}{j}_1 + \underset{\sim}{j}_2 + \underset{\sim}{j}_3 + \ldots = \underset{\sim}{J}$.

This scheme is appropriate for heavy atoms, where relativistic effects and the spin- or b.f interaction are significant.

Despite this, terms are usually defined through the LS coupling scheme.

Units and database — energies

NIST database, https://physics.nist.gov/PhysRefData/ASD/levels_form.html .

Also useful reference, for energy conversion: https://physics.nist.gov/cuu/Constants/energy.html .

Coupling of angular momentum

Clebsch-Gordan coefficients and 3-j symbols

Clebsch-Gordan coefficients are the coefficients that relate eigenfunctions defined by one set of operators to those defined by another set of operators.

E.g. $|j,m\rangle = \sum_{m_1,m_2} C(j_1,j_2,j\,;m_1,m_2,m)\,|j_1,m_1\rangle\,|j_2,m_2\rangle$.

- $|j,m\rangle$: state that is an eigenfunction of J^2 and J_z,
  - $J^2|j,m\rangle = j(j+1)|j,m\rangle$
  - $J_z|j,m\rangle = m|j,m\rangle$.
- $\sum_{m_1,m_2}$: sum over all m_1 and m_2, so no dependence on these explicitly in the new state.
- $C(j_1,j_2,j\,;m_1,m_2,m)$: C-G coefficients are the expansion coefficients
- $|j_1,m_1\rangle\,|j_2,m_2\rangle$: These are eigenfunctions of J_1^2, J_{1z} and J_2^2, J_{2z}, respectively.

E.g. $J_1^2|j_1,m_1\rangle = j_1(j_1+1)|j_1,m_1\rangle$
$J_{1z}|j_1,m_1\rangle = m_1|j_1,m_1\rangle$

More explicit symmetry in angular momentum coupling when a new construct is introduced:

3-j symbol

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1 - j_2 - m_3}}{\sqrt{2j_3+1}} \times C(j_1, j_2, j_3; m_1, m_2, m_3).$$

Some properties of 3-j symbols:

- projections must add to zero:
$$m_1 + m_2 + m_3 = 0.$$
- symmetric on even permutations:

e.g. $$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \begin{pmatrix} j_2 & j_3 & j_1 \\ m_2 & m_3 & m_1 \end{pmatrix}$$

- Phase introduced on odd permutations:

e.g. $\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_3 & j_2 \\ m_1 & m_3 & m_2 \end{pmatrix}$

- Phase introduced when changing the sign of all m:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}$$

And this brings us to the Wigner-Eckart theorem and the spherical tensor formalism.

Wigner-Eckart theorem

Matrix element.

$$\langle j_1, m_1 | T^k_q | j_2, m_2 \rangle$$

final state. — operator of rank k, projection q. — initial state

T^k_q — spherical tensor operator, irreducible

Operator & wavefunctions on same footing.

$$\langle j_1 m_1 | T^k_q | j_2 m_2 \rangle$$

reduced matrix element

$$= (-1)^{j_1 - m_1} \begin{pmatrix} j_1 & k & j_2 \\ -m_1 & q & m_2 \end{pmatrix} \langle j_1 \| T^k \| j_2 \rangle$$

3-j symbol

$$j_2 = j_1 + k, j_1 + k - 1, \ldots |j_1 - k|$$

T^k_q is a spherical irreducible tensor operator if

$$[J_z, T^k_q] = q\, T^k_q$$

$$[J_\pm, T^k_q] = [(k \pm q + 1)(k \mp q)]^{1/2}\, T^k_{q \pm 1}$$

Exercise: Determine

$$\langle l_1 \| C^k \| l_2 \rangle,$$

where $C^k_q = \sqrt{\frac{4\pi}{2k+1}}\, Y_{kq}(\theta, \varphi)$

↑ irreducible tensor operator

↑ spherical harmonic

Spherical tensor operators

$T_{ij} = a_i b_j$

For use in Wigner–Eckart thm, must be irreducible spherical tensor operators.

In general, T_{ij} is reducible.

$$T_{ij} = a_i b_j = \underbrace{\frac{\underset{\sim}{a}\cdot\underset{\sim}{b}}{3}\delta_{ij}}_{\text{scalar } T^{(0)}_{ij}} + \underbrace{\frac{a_i b_j - a_j b_i}{2}}_{\text{vector } T^{(1)}_{ij}\ [\underset{\sim}{a}\times\underset{\sim}{b}]} + \underbrace{\left(\frac{a_i b_j + a_j b_i}{2} - \frac{\underset{\sim}{a}\cdot\underset{\sim}{b}}{3}\delta_{ij}\right)}_{\text{2nd-rank tensor } T^{(2)}_{ij}}$$

Each of these cannot be further decomposed.

components in total

T_{ij} $3 \times 3 = 9$.

Broken into irreducible tensors, each tensor contributes $2K+1$ components, where K is the rank.

Scalar: $2(0)+1 \Rightarrow 1$.

Vector: $2(1)+1 \Rightarrow 3$

2nd-rank tensor: $2(2)+1 \Rightarrow 5$

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So far, this is a Cartesian treatment. We want spherical tensor operators.

Cartesian → spherical.

x, y, z

Vector

$$V_1 = -\frac{1}{\sqrt{2}}(V_x + iV_y)$$

$$V_0 = V_z$$

$$V_{-1} = \frac{1}{\sqrt{2}}(V_x - iV_y).$$

Unit vectors:

$$\hat{e}_1 = -\frac{1}{\sqrt{2}}(\hat{e}_x + i\hat{e}_y)$$

$$\hat{e}_0 = \hat{e}_z$$

$$\hat{e}_{-1} = \frac{1}{\sqrt{2}}(\hat{e}_x - i\hat{e}_y).$$

$\hat{e}^*_{\pm 1} = -\hat{e}_{\mp 1}$

$\underset{\sim}{u} = \sum_q u_q \hat{e}^*_q$

$\hat{e}^*_q \cdot \hat{e}_{q'} = \delta_{qq'}$

⇒ scalar product of vectors:

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = -u_1 v_{-1} + u_0 v_0 - u_{-1} v_1$$

$$= \sum_q (-1)^q u_q v_{-q}.$$

⇒ exercise for you

Tensor (spherical) scalar product:

$$T^{(k)} \cdot U^{(k)} = \sum_q (-1)^q T^k_q U^k_{-q}$$

Coming back to spherical tensors (irreducible) from cartesian tensors:

$$T^2_0 = T^{(2)}_{zz}$$

$$T^2_{\pm 1} = \mp\sqrt{\frac{2}{3}}\left(T^{(2)}_{zx} \pm i T^{(2)}_{zy}\right)$$

$$T^2_{\pm 2} = \sqrt{\frac{1}{6}}\left(T^{(2)}_{xx} - T^{(2)}_{yy} \pm 2i T^{(2)}_{xy}\right)$$

Exercise:

Show that:

$$\langle \gamma\, I\, J\, F\, M \,|\, (T^k U^k) \,|\, \gamma'\, I'\, J'\, F'\, M' \rangle =$$

$$= (-1)^{I'+J+F} \sum_{\gamma''} \langle \gamma I \| T^k \| \gamma'' I' \rangle \langle \gamma'' J \| U^k \| \gamma' J' \rangle \times$$

$$\begin{Bmatrix} I & J & F \\ J' & I' & k \end{Bmatrix}$$

Hartree - Fock.

Total Hamiltonian for N electrons:

$$H(\underset{\sim}{r}_1, \underset{\sim}{r}_2, \ldots, \underset{\sim}{r}_N) = \sum_{i=1}^{N} h_0(\underset{\sim}{r}_i) + \sum_{i<j} \frac{1}{|\underset{\sim}{r}_i - \underset{\sim}{r}_j|}$$

h_0 – single-particle operator,

$$h_0(\underset{\sim}{r}) = -\frac{1}{2}\nabla^2 - \frac{Z}{r}$$

Seek solutions:

$$H\Psi = E\Psi$$

↑

Many-electron wavefunction.

⇒ needs to be antisymmetric, etc.

⇒ suggested simple solution given by Slater determinant.

$$\psi_{ab\ldots n}(\underset{\sim}{r}_1, \underset{\sim}{r}_2, \ldots \underset{\sim}{r}_N)$$

$$= \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_a(\underset{\sim}{r}_1) & \psi_b(\underset{\sim}{r}_1) & \cdots & \psi_n(\underset{\sim}{r}_1) \\ \psi_a(\underset{\sim}{r}_2) & & & \\ \vdots & & & \\ \psi_a(\underset{\sim}{r}_N) & \psi_b(\underset{\sim}{r}_N) & \cdots & \psi_n(\underset{\sim}{r}_N) \end{vmatrix}$$

We break up the Hamiltonian into two parts:

1st part:

$$H_0(\underset{\sim}{r}_1, \underset{\sim}{r}_2, \ldots, \underset{\sim}{r}_N) = \sum_{i=1}^{N} h(\underset{\sim}{r}_i)$$

single-particle Hamiltonian

$$= \sum_{i=1}^{N} \left[h_0(\underset{\sim}{r}_i) + U(\underset{\sim}{r}_i) \right]$$

2nd part:

$$V(\underset{\sim}{r}_1, \underset{\sim}{r}_2, \ldots \underset{\sim}{r}_N) = \sum_{i<j} \frac{1}{|\underset{\sim}{r}_i - \underset{\sim}{r}_j|} - \sum_i U(\underset{\sim}{r}_i)$$

$$H = H_0 + V$$

$\Rightarrow$ "Hartree - Fock" equations (with explicit expression for potential U) obtained by taking expectation value of Hamiltonian with Slater determinant.

Note: Slater determinant formed from orbitals that are solutions of H_0.

How do we take matrix element of 1- & 2- particle operators with N-particle wave function?

single-particle operator.

↓

$$\langle \psi_{a'b'c'\ldots n'} | F | \psi_{abc\ldots n} \rangle$$

↑

many-electron wavefunction

$$F = \sum_i^N f(\hat{r}_i)$$

If ψ different by more than 1 particle. $\Rightarrow \langle \psi' | F | \psi \rangle = 0.$

If ψ diff. by 1 index:

$$\langle \psi_{ab\ldots k'\ldots} | F | \psi_{ab\ldots k\ldots} \rangle = f_{k'k}$$

If identical:

$$\Rightarrow \langle \psi_{ab\ldots} | F | \psi_{ab\ldots} \rangle = \sum_{i=a}^{n} f_{ii}$$

Similarly for 2-particle operator $G = \sum_{i<j} g(r_{ij})$,

though more complicated!

$$\left[r_{ij} = |\underset{\sim}{r}_i - \underset{\sim}{r}_j| \right]$$

If they differ by 2 indices:

$$\langle \psi_{ab\ldots k'\ldots l'\ldots} | G | \psi_{ab\ldots k\ldots l} \rangle$$

$$= g_{k'l'kl} - g_{k'l'lk}$$

If differ by 1:

$$\langle \psi_{ab\ldots k'\ldots} | G | \psi_{ab\ldots k\ldots} \rangle$$

$$= \sum_i \left(g_{k'ika} - g_{k'iik} \right)$$

If identical:

$$\langle \psi | S | \psi \rangle = \frac{1}{2} \sum_{i,j} \left(g_{ijij} - g_{jiij} \right)$$

$$f_{ab} = \langle a | f | b \rangle$$

$$= \int d^3 r_1 \, \psi_a^\dagger(\underset{\sim}{r}_1) \, f(\underset{\sim}{r}_1) \, \psi_b(\underset{\sim}{r}_1) .$$

$$g_{abcd} = \langle ab | g | cd \rangle$$

$$= \int d^3 r_1 \int d^3 r_2 \, \psi_a^\dagger(\underset{\sim}{r}_1) \psi_b^\dagger(\underset{\sim}{r}_2)$$

$$g(r_{12}) \, \psi_c(\underset{\sim}{r}_1) \psi_d(\underset{\sim}{r}_1) .$$

Remainder of procedure for finding Hartree-Fock equations:

Introduce Lagrange multipliers, & require that the energy is stationary on variation of radial wavefunctions:

Eventually, obtain:

$$-\frac{1}{2}\frac{d^2P_a}{dr^2} + \left(V_{HF} - \frac{Z}{r} + \frac{l_a(l_a+1)}{2r^2}\right)P_a(r) = \varepsilon_a P_a(r)$$

to be solved self-consistently for each orbital P_a.

$$V_{HF} = V_{dir}\,P(r) + V_{exch}\,P(r)$$

$$= \sum_b (4l_b+2)\,N_0(b,r)\,P_a(r) + \sum_b (4l_b+2) \sum_l \Lambda_{l_b l l_a}\, N_l(b,a,r)\,P_b(r)$$

See Johnson, eqns. (3.75, 3.76), and preceding equations for notations.

Origin of Vcell? Antisymmetry of wavefunctions!

Dirac equation : reduction to angular form.

{Johnson, p. 51}.

Relativistic effects and fine structure included from the start.

1 electron in the field of a central potential $V(r)$:

$$h_D \varphi = E \varphi,$$

$$h_D = c \underset{\sim}{\alpha} \cdot \underset{\sim}{p} + \beta c^2 + V(r).$$

$\underset{\sim}{\alpha}$, β are Dirac matrices, $\underset{\sim}{\alpha} = \begin{pmatrix} 0 & \underset{\sim}{\sigma} \\ \underset{\sim}{\sigma} & 0 \end{pmatrix}$, $\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$,

with $\underset{\sim}{\sigma}$ the Pauli spin matrix.

Here, $E = \mathcal{E} + mc^2$

($\mathcal{E}$: binding energy; mc^2: rest mass)

It is quite common to solve directly for the binding energies, with

$$h'_D = c \underset{\sim}{\alpha} \cdot \underset{\sim}{p} + (\beta - 1) c^2 + V(r).$$

Though here we follow Johnson.

Solutions of the Dirac equation contain two radial components — the "upper" and "lower" or "big" and "small" radial wave functions, called P and Q in Johnson.

They also contain a new angular wave function — $\Omega_{\kappa m}$ or Ω_{jlm} — which is made up from a combination of spherical harmonics and spinors and is called a spherical spinor.

Solution of the Dirac equation takes the form

$$\varphi_{n\kappa m}(\vec{r}) = \frac{1}{r}\begin{pmatrix} i P_{n\kappa}(r)\,\Omega_{\kappa m}(\hat{r}) \\ Q_{n\kappa}(r)\,\Omega_{-\kappa m}(\hat{r}) \end{pmatrix}.$$

Things to note:

- Q is smaller than P, and it is quite common to pull out a factor of $\alpha \approx \frac{1}{137}$ from Q to make them

more comparable.

- $\Omega_{-\kappa m} = (-\underset{\sim}{\sigma}\cdot\hat{\underset{\sim}{r}})\,\Omega_{\kappa m}$

  (and often denoted $\tilde{\Omega}_{\kappa m}$).

- κ is the relativistic angular momentum quantum number. It is the eigenvalue of the operator $K = -1 - \underset{\sim}{\sigma}\cdot\underset{\sim}{L}$.

So $K\,\Omega_{\kappa m}(\hat{\underset{\sim}{r}}) = \kappa\,\Omega_{\kappa m}(\hat{\underset{\sim}{r}})$.

It includes the spin-orbit coupling

κ takes the values

$$\kappa = \mp(j+\tfrac{1}{2}) \quad \text{for} \quad j = \ell \pm 1/2 .$$

This corresponds to:

κ =	-1,	1,	-2,	2,	-3,	3,	-4,	...
ℓ_j =	$s_{1/2}$,	$p_{1/2}$,	$p_{3/2}$,	$d_{3/2}$,	$d_{5/2}$,	$f_{5/2}$,	$f_{7/2}$,	...

- the lower angular component $\Omega_{-\kappa m}$ has the same angular momentum j though opposite parity (determined by ℓ) c.f. the upper angular component $\Omega_{\kappa m}$.

The Dirac equation may be readily reduced to two coupled equations for the radial wave functions:

$$\left(V + c^2\right) P_\kappa + c\left(\frac{d}{dr} - \frac{\kappa}{r}\right) Q_\kappa = E P_\kappa$$

$$-c\left(\frac{d}{dr} + \frac{\kappa}{r}\right) P_\kappa + \left(V - c^2\right) Q_\kappa = E Q_\kappa .$$

The normalisation of $\varphi(\vec{r})$ leads to

$$\int_0^\infty dr \left(P_\kappa^2(r) + Q_\kappa^2(r)\right) = 1.$$

&

$$\langle \Omega_{\kappa' m'} | \Omega_{\kappa m} \rangle = \delta_{\kappa' \kappa} \delta_{m' m}.$$

There are some things to be mindful of, & a couple of nice tricks, in deriving this.

- be mindful of the ordering in the operators and wave functions.
- nice trick:

manipulate this

$$\underset{\sim}{\sigma}\cdot\underset{\sim}{p} = \underbrace{(\underset{\sim}{\sigma}\cdot\underset{\sim}{n})(\underset{\sim}{\sigma}\cdot\underset{\sim}{n})}_{=1}\,\underset{\sim}{\sigma}\cdot\underset{\sim}{p}$$

Use $(\underset{\sim}{\sigma}\cdot\underset{\sim}{A})(\underset{\sim}{\sigma}\cdot\underset{\sim}{B}) = \underset{\sim}{A}\cdot\underset{\sim}{B} + i\,\underset{\sim}{\sigma}\cdot(\underset{\sim}{A}\times\underset{\sim}{B})$

$$\Rightarrow \quad \underset{\sim}{\sigma}\cdot\underset{\sim}{p} = \underset{\sim}{\sigma}\cdot\underset{\sim}{n}\left[\underset{\sim}{n}\cdot\underset{\sim}{p} + i\,\underset{\sim}{\sigma}\cdot(\underset{\sim}{n}\times\underset{\sim}{p})\right]$$

$$= -i\,\underset{\sim}{\sigma}\cdot\underset{\sim}{n}\left[i\,\underset{\sim}{n}\cdot\underset{\sim}{p} - \underset{\sim}{\sigma}\cdot\frac{(\underset{\sim}{r}\times\underset{\sim}{p})}{r}\right] \qquad \underset{\sim}{L}$$

$$= -i\,\underset{\sim}{\sigma}\cdot\underset{\sim}{n}\left[\frac{d}{dr} - \frac{1}{r}\,\underset{\sim}{\sigma}\cdot\underset{\sim}{L}\right]$$

$$= -i\,\underset{\sim}{\sigma}\cdot\underset{\sim}{n}\left[\frac{d}{dr} + \frac{1}{r}(K+1)\right]$$

$$\Rightarrow \hat{\sigma}\cdot\hat{r}\, f(r)\,\Omega_{\kappa m}$$

$$= -i\,\hat{\sigma}\cdot\hat{r}\left(\frac{df}{dr} + \frac{\kappa+1}{r}\right)\Omega_{\kappa m}$$

$$= i\left(\frac{df}{dr} + \frac{\kappa+1}{r}\right)\Omega_{-\kappa m}.$$

Exercise: complete the derivation of the coupled radial Dirac equations.

Exercise: use the solutions P & Q in the Mathematica notebook to calculate matrix elements $\left\langle \frac{1}{r^2} \right\rangle$ and $\langle r \rangle$ and compare to the non-relativistic results.

After these lectures, you should understand

- the general structure of atomic levels
- how electron angular momenta couple, and the meaning of the term designation
- the meaning of Wigner-Eckart theorem, and how to use 3-j and 6-j symbols, and spherical tensor operators
- what the Hartree-Fock approximation is
- how the coupled radial Dirac equation appears, and how to go about evaluating simple matrix elements